ALGEBRA 2 · UNIT 1: FUNCTIONS & TRANSFORMATIONS

Ride-Share Pricing Lab

Model ride fares with piecewise functions, transformations, composition, and inverses.

Honors

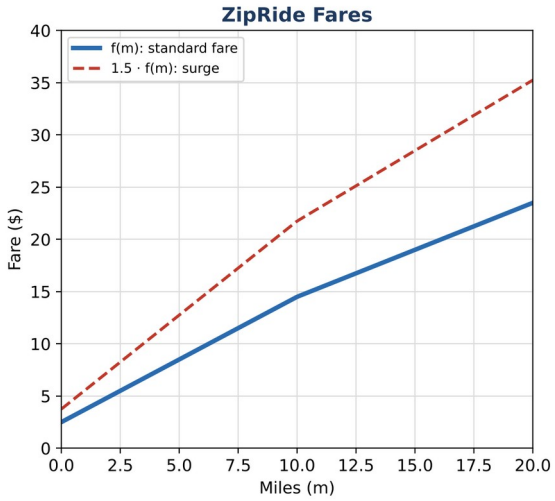
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

ZipRide is a new ride-share app in town. Its fare depends on miles traveled, but the rate changes after 10 miles, during busy times there is a surge multiplier, and the app is considering a booking fee.

Every one of these rules is a function idea: piecewise definitions, transformations, composition, and inverses. You will analyze ZipRide and then design a plan for a competitor.



Fare Rules

Base fare $2.50

First 10 miles:

$1.20 per mile

After 10 miles:

$0.90 per mile

Surge: × 1.5

Learning Goals

- I can analyze transformations and their order.
- I can find inverses of composed and piecewise functions with domains.
- I can design a function that meets several conditions.

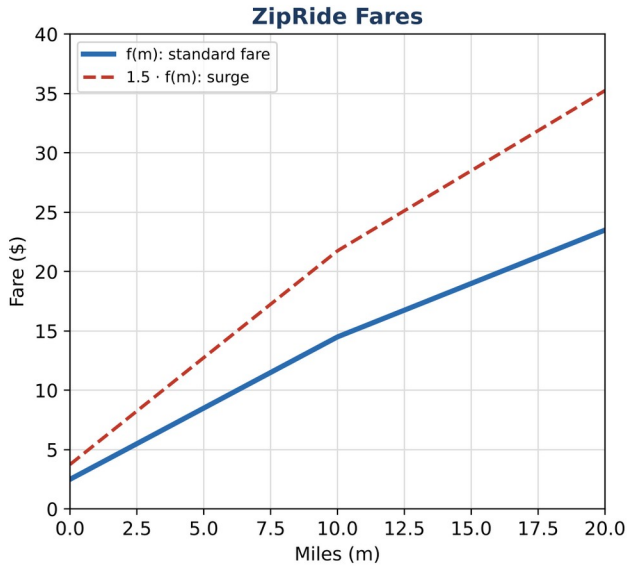
Project Timeline

When	What you will do
Day 1	Fare Functions
Day 2	Composition & Inverse
Day 3	Engineer a Plan
Final	Pricing Plan Pitch + reflection

Part 1: Fare Functions

Day 1 · Use the Fare Rules card.

1. Write f(m) piecewise and find its domain and range. Is f one-to-one? Explain.



Fare Rules

Base fare $2.50

First 10 miles:

$1.20 per mile

After 10 miles:

$0.90 per mile

Surge: × 1.5

2. Compare $F_1(m) = 1.5(f(m) + 2)$ and $F_2(m) = 1.5 \cdot f(m) + 2$. Which is fair to riders, and how much do they differ?

3. Describe F_1 as a sequence of transformations of f, in order.

Part 2: Composition & Inverse

Day 2 · Exact rules with domains.

4. Traffic: $m(t) = 0.4t$ for $0 \le t \le 15$ and $m(t) = 6 + 0.8(t - 15)$ for $t > 15$. Write $f(m(t))$ as a piecewise function of t.

5. Find the inverse of $f(m(t))$ and use it to find the ride time for a \$19.00 fare.

6. Prove that $(f^{-1} \circ f)(m) = m$ for $m > 10$.

Part 3: Engineer a Plan

Day 3 · Conditions: continuous; base fare between $2 and $4; the 8-mile fare is $11; the 20-mile fare is $20.

7. Design a two-piece plan that meets every condition. Show your algebra.

8. Find where your plan and ZipRide cost the same.

9. Write the inverse and explain one marketing use.

Final Product: Pricing Plan Pitch

Your final product must include:

- Piecewise analysis and transformation comparison
- Composed piecewise function and its inverse
- Engineered plan meeting all conditions
- Pitch

Reflection

R1. Why does the order of a fee and a surge matter?

R2. How do domains of the pieces become ranges of the inverse pieces?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Pricing plan is a correct piecewise function with a correct inverse or composition used to answer a customer question. Goes beyond the minimum with a creative, realistic choice.	Pricing plan is a correct piecewise function with a correct inverse or composition used to answer a customer question.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Pitch shows the piecewise fare function, its graph, transformations for fees or surges, and a comparison with ZipRide. Polished, organized, and visually engaging.	Pitch shows the piecewise fare function, its graph, transformations for fees or surges, and a comparison with ZipRide. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.